



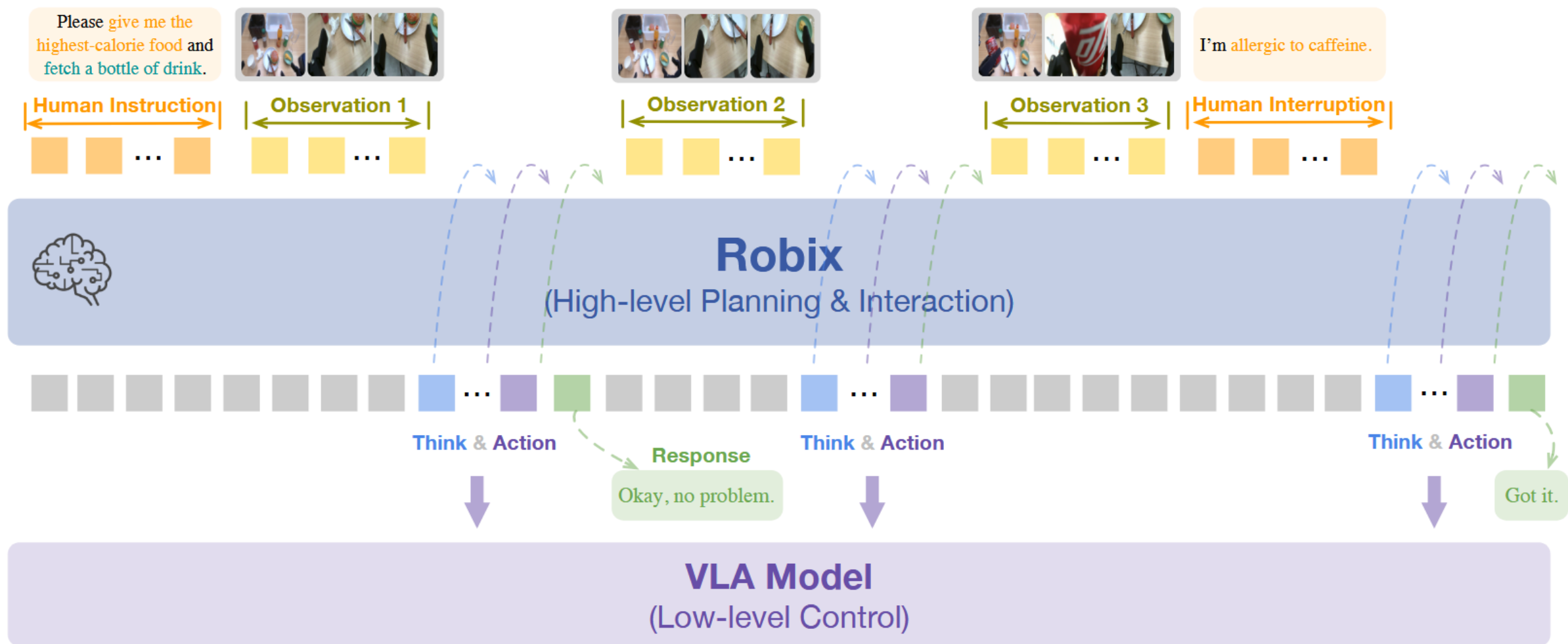
西北工业大学
NORTHWESTERN POLYTECHNICAL UNIVERSITY

Understanding Prediction Execution

Zhengwei Yang

2026.05.31

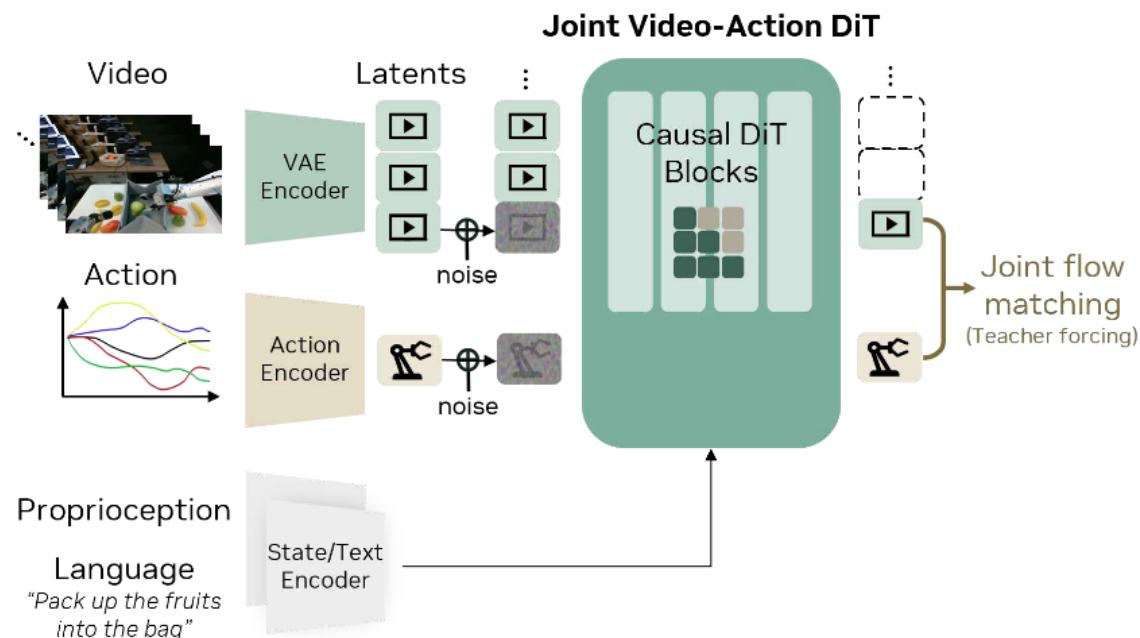
Robix: A Unified Model for Robot Interaction, Reasoning and Planning



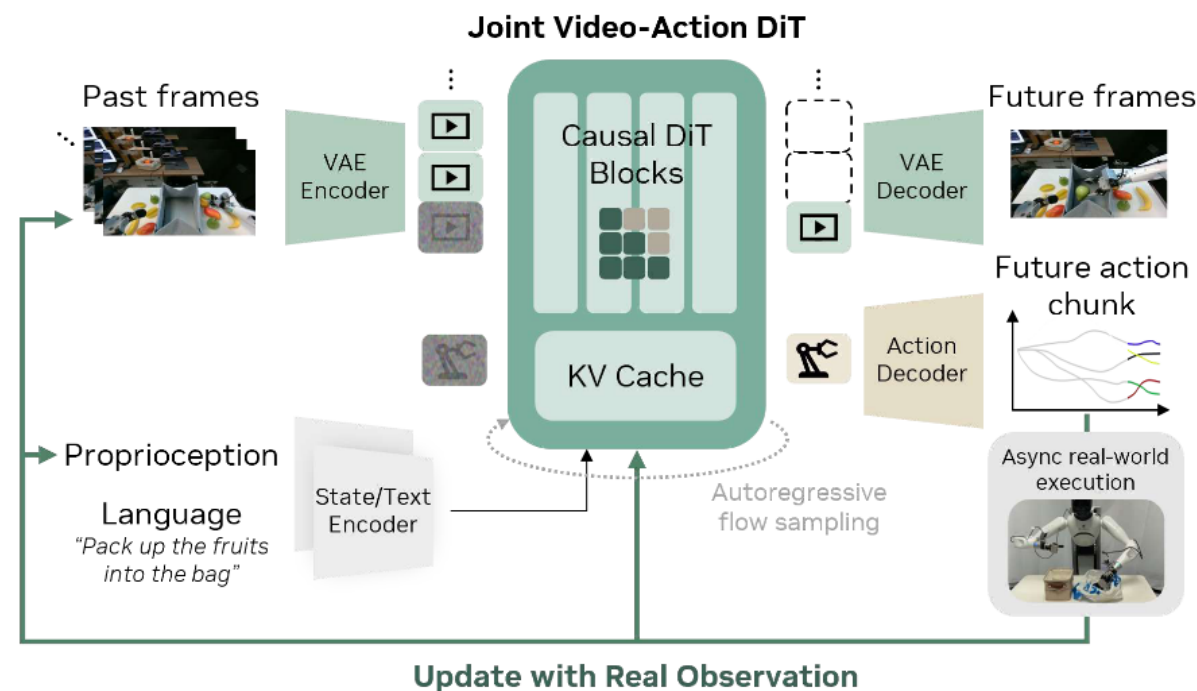
- CoT Reasoning
- Three-stage training strategy: Continued pretraining → SFT → RL

World Action Models are **Zero-shot** Policies

Training: Joint Video-Action Flow Matching



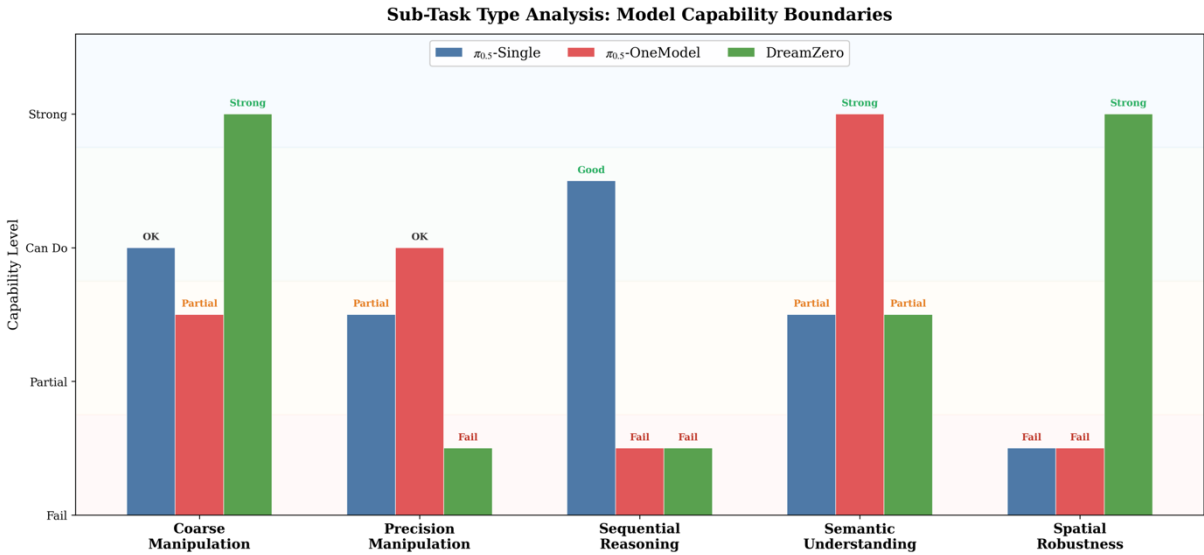
Inference: Closed-Loop Real World Execution



$$\underbrace{\pi_0(\mathbf{o}_{l:l+H}, \mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{DREAMZERO}} = \underbrace{\pi_0(\mathbf{o}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{video prediction}} \underbrace{\pi_0(\mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l+H}, \mathbf{q}_l)}_{\text{IDM}}$$

Motivation

Category	Task	$\pi_{0.5}$ -Single		$\pi_{0.5}$ -OneModel		DreamZero	
		Score	SR	Score	SR	Score	SR
Execution	arrange_cup	18	0%	14	0%	25	0%
	put_spoon	52	20%	43	10%	54	20%
	put_glasses	80	60%	87	70%	37	0%
	put_ring	41	20%	60	20%	27	0%
	put_items_drawer	34	0%	7	0%	7	0%
	pick_items_basket	46.8	20%	37	10%	97.8	90%
	pour_water	8	0%	19	0%	12	0%
	insert_wireline	24	0%	21	0%	24	0%
	put_stationery	21.5	0%	16	0%	26	0%
	put_blocks	48	30%	42	30%	27	0%
Semantic	sort_headphone	35	20%	73	70%	66	60%
	classify_items	59	20%	63	40%	36	0%
	press_button	48	20%	13	0%	3	0%
	pair_up_items	21	0%	51.5	30%	13.5	0%
	pick_fruits	90	80%	94	80%	45	0%
Mobile Manipulation (5 tasks): evaluation in progress, to be reported in subsequent update.							
Tabletop Total (/ 1,500)		626.3		640.5		500.3	



CVPR 2026 ManipArena Challenge Results

Pros (+)
Cons (-)



Linguistic
Planning



LLM-based

- Strong logical reasoning
- Rich common-sense knowledge
- Strong cross-task transfer capability
- Minimal computational overhead and low latency
- Lack of physical and spatial-geometric awareness
- Poor spatial generalization ability



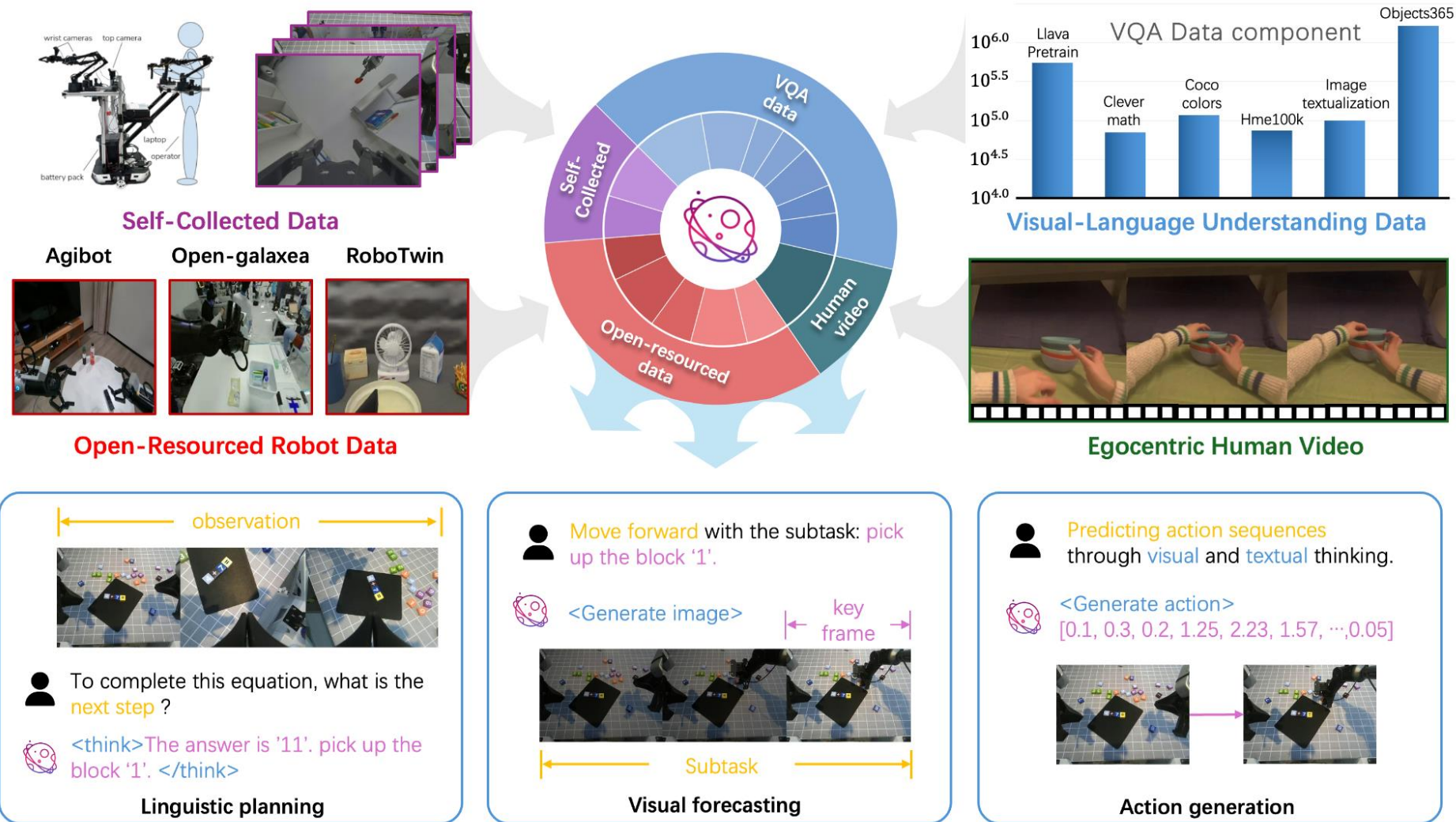
Visual
Forecasting



Video Prediction Policy

- Strong physical consistency and spatial robustness
- Excel at coarse/basic tasks and manipulation
- Latent ability of causal reasoning
- Action-state alignment
- Failed to complete fine-grained tasks
- High latency and huge computational overhead

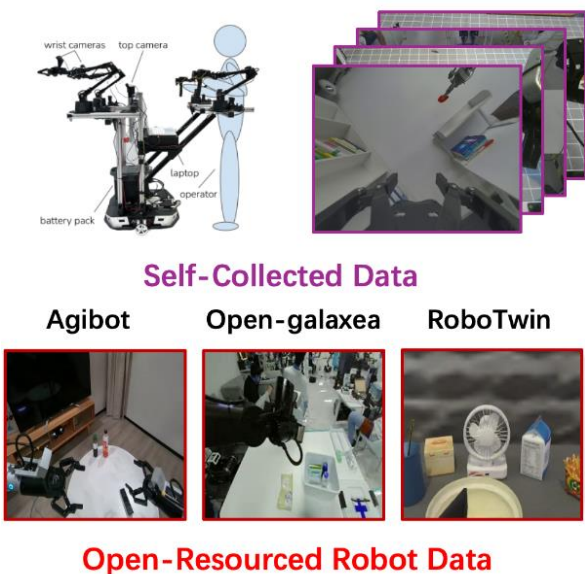
BagelVLA: Enhancing Long-Horizon Manipulation via **Interleaved** Vision-Language-Action Generation



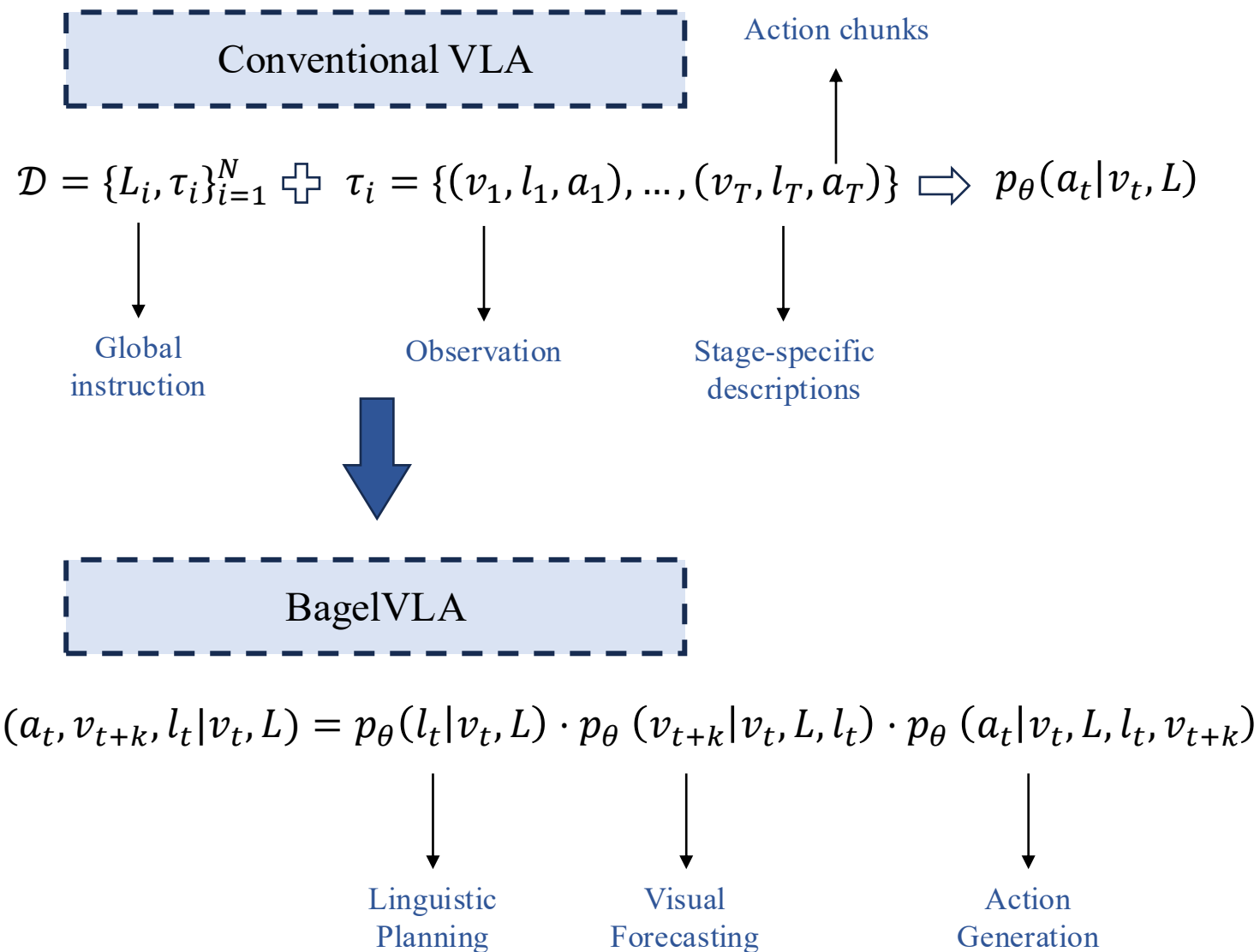
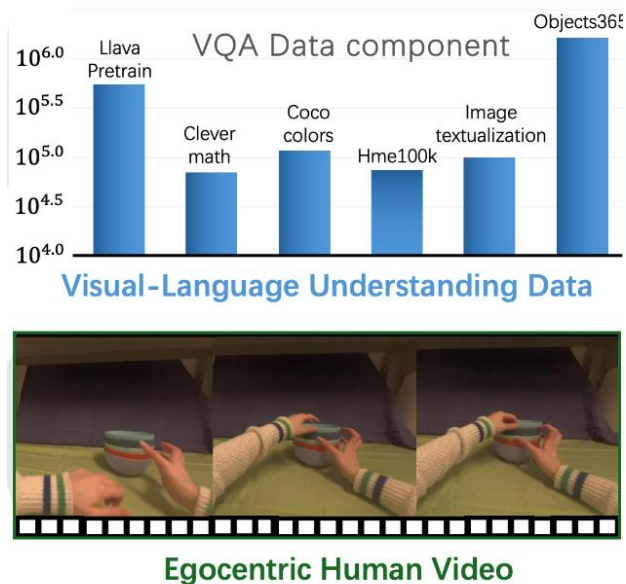
Overview of BagelVLA

Dataset

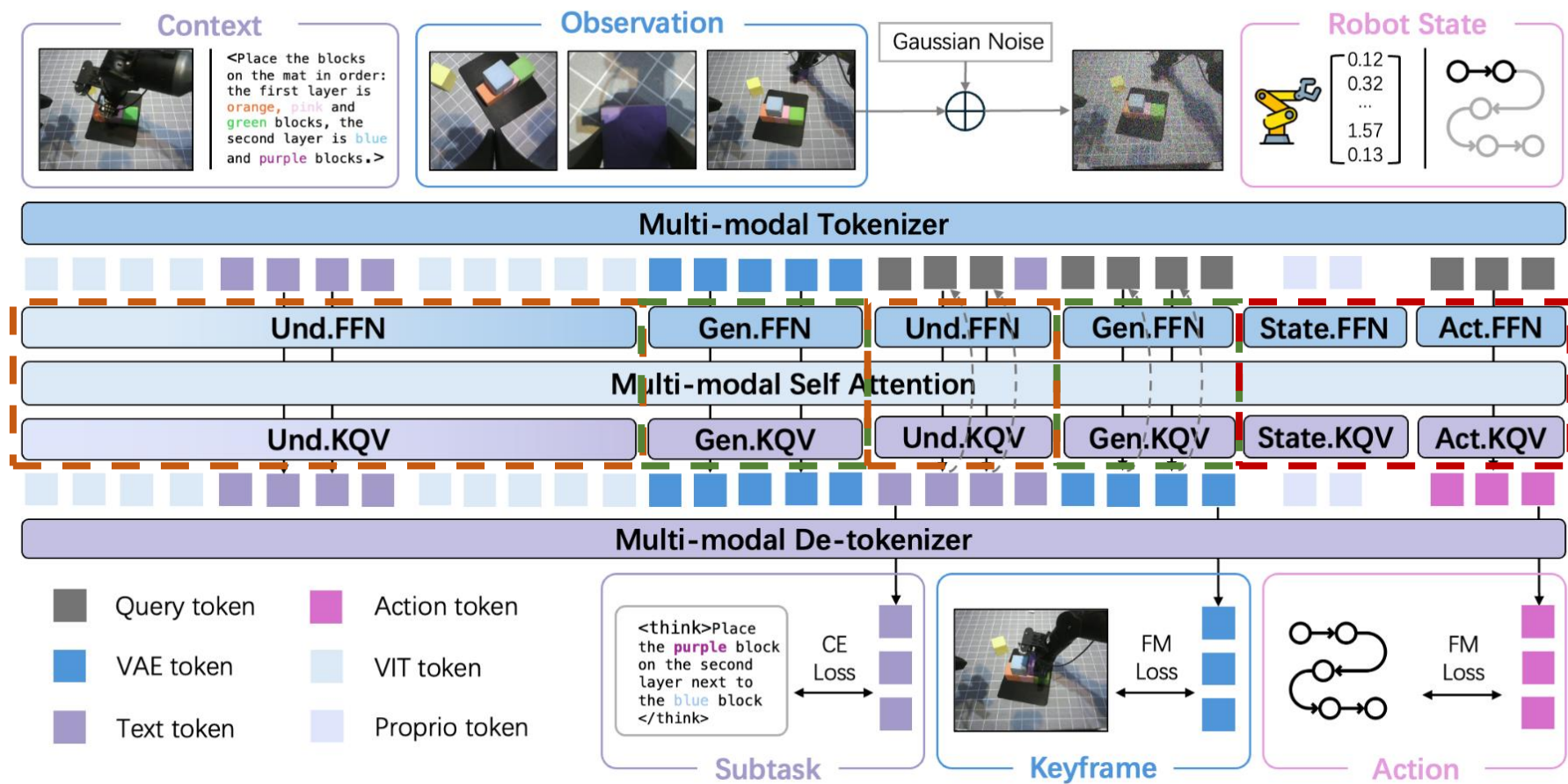
Large-scale Robotic Datasets



General Multimodal Data

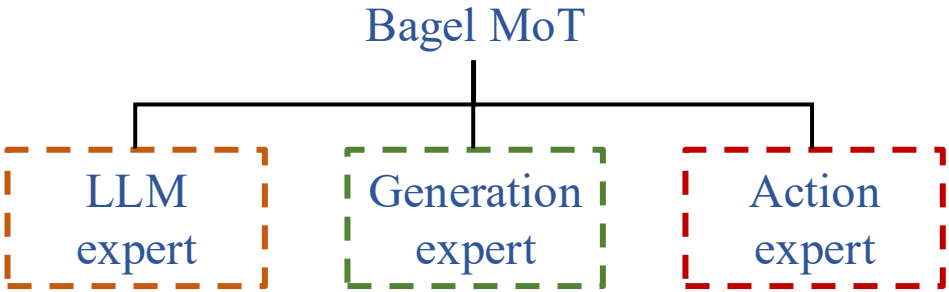


Architecture



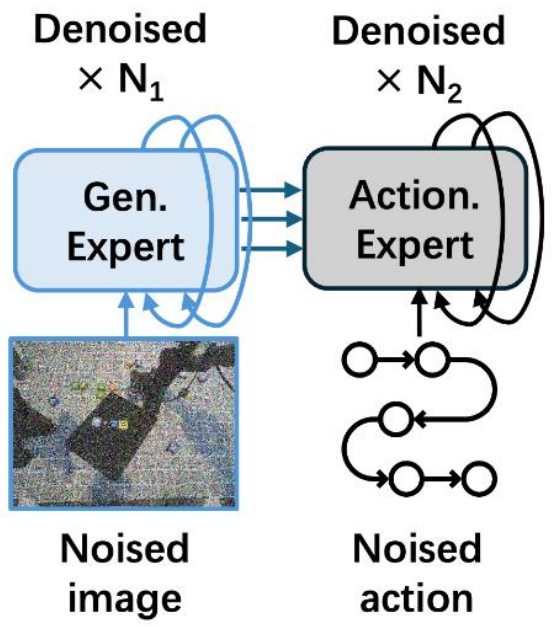
Modules	Understanding Expert	Generation Expert	Action Expert
Size	7B	7B	2B
Input Modality	Image/Text	Image	Proprio/Action
Output Modality	Text	Image	Action
Encoder	ViT+MLP	VAE+MLP	MLP
Image Resolution	256x256	256x256 (VAE)	-
Hidden size	3584	3584	3584
Intermediate size	18944	18944	3584
Layers	28	28	28
Loss Type	CE	MSE(FM)	MSE(FM)
FM Timestep Distribution	-	LogitNormal(0, 1)	Beta(1.5,1)

* Designed for higher execution frequency

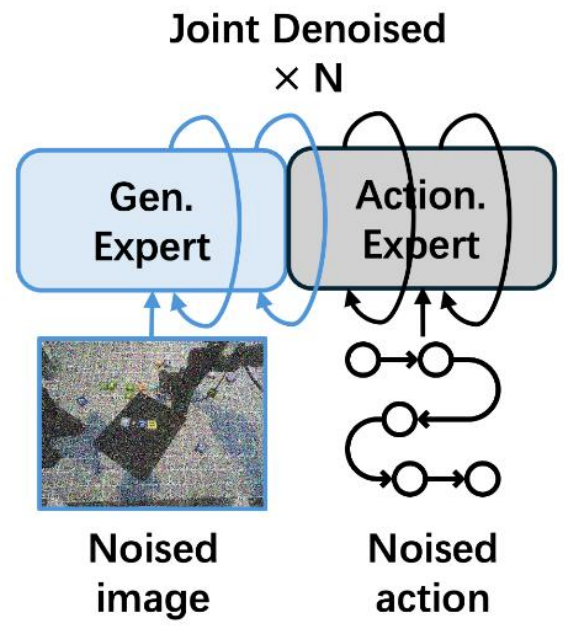


Dual Flow-Matching

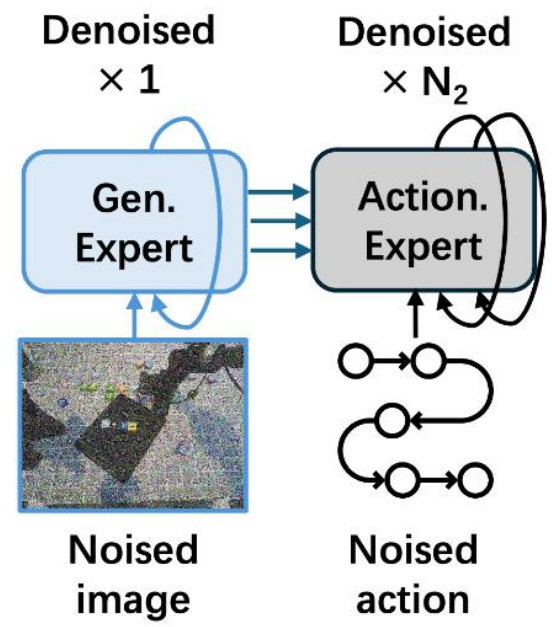
Pure noise image?
Noised image?



a. Complete Denoise



b. Joint Denoise



c. Single-Step Denoise

$$v_{t+k}^{\tau=0} \rightarrow v_{t+k}^{\tau=1} = v_{t+k}$$

$$a_t \mid v_{t+k}$$

~ DreamZero
WM + IDM

- Based on “a realistic future”
- Inference latency: $N_1 + N_2$ steps
- Accumulation of visual errors

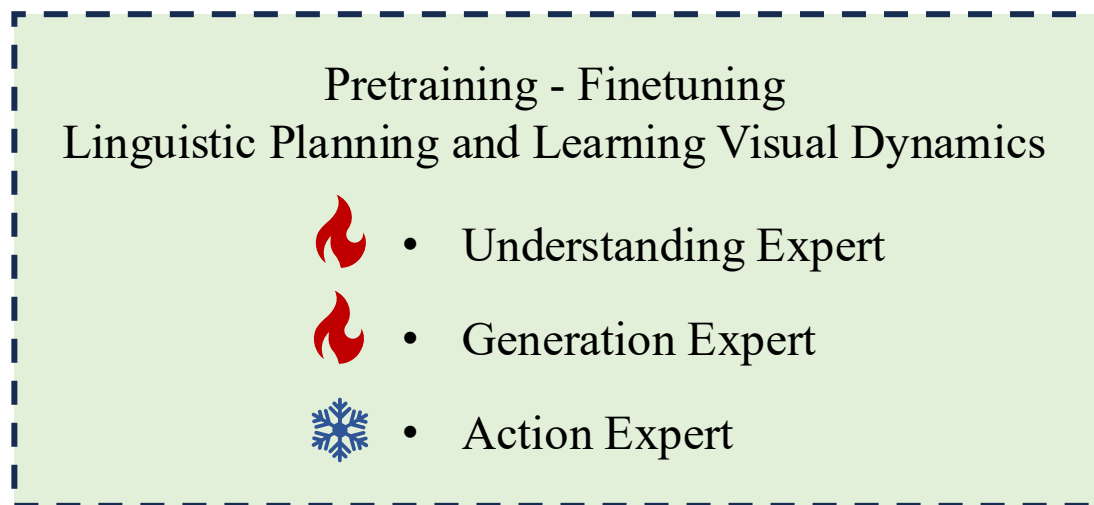
$$a_t^\tau \mid v_{t+k}^\tau$$

- Faster inference: N steps
- Spurious correlation

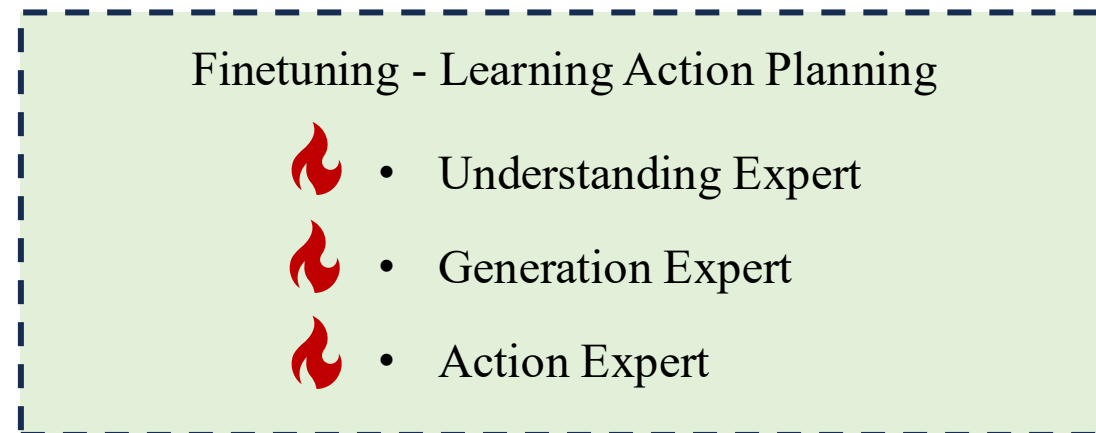
$$a_t^\tau \mid v_{t+k}^{\tau=0} \quad \begin{matrix} \text{X} & v_{t+k}^{\tau=0} \sim \mathcal{N}(0, I) \\ \text{RFG} \rightarrow \text{checkmark} & v_{t+k}^{\tau=0} \sim \mathcal{N}(v_t, I) \end{matrix}$$

- Fastest inference
- focus on modeling robot manipulation changes

Training



Acquire capabilities in
textual planning and keyframe prediction



Perform robustly in specific scenarios

Trick: Randomly replace the current frame with a preceding image

48Hz

Inference



72Hz

Asynchronous Execution

Reduce the updating frequency of the KV contexts of
understanding and generation experts

At each denoising step, only a single expert is activated

Single-step denoise scheme

Results

Average Completion Length		Success Rate	
Model	Calvin	Robotwin	
	ABC-D	Clean	Randomized
π_0	3.648	46.42	16.34
RDT	-	34.50	13.72
UP-VLA	4.078	52.92	15.16
VPP	4.329	-	-
w/o Textual-planning	-	54.00	19.20
w/o Keyframe-forecasting	3.345	56.72	15.92
BagelVLA	4.405	75.26	20.87

Simulation

Model	Pick&Place Seen	Pick&Place Unseen	Water Flower	Stack Cubes	Put Flowers in Vase	Stack Bowls	Pour Fries	Sweep Rubbish	Press Button	Drawer Close	Success Average
π_0	95	55	50	65	40	70	35	55	90	95	65.0
VPP	85	45	60	50	50	55	30	45	75	100	59.5
BagelVLA	95	85	60	80	35	90	45	80	90	95	75.5

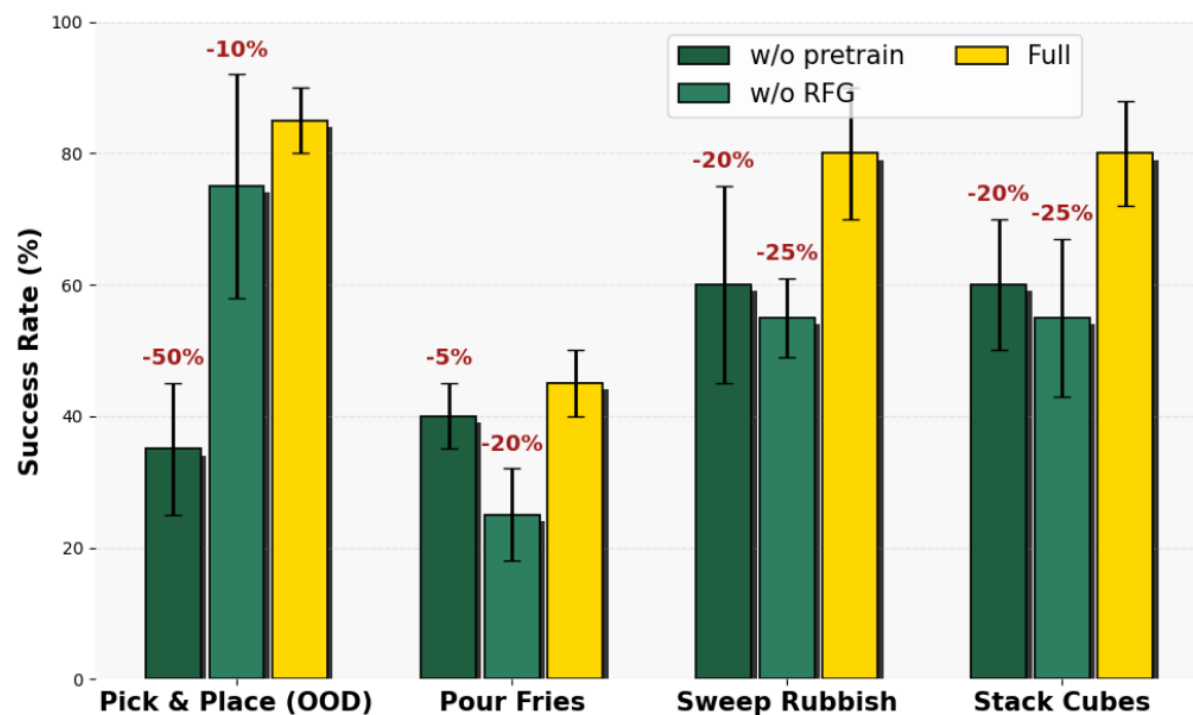
Basic Task

Tasks		Stack Cubes in Requested Order					Calculate and Place Symbol Blocks				
Difficulty		Easy	Middle	Hard	Success Rate↑	Planning Accuracy↑	Easy	Middle	Hard	Success Rate↑	Planning Accuracy↑
π_0		75	35	10	40.0	55	70	25	0	31.7	40
VPP		60	15	0	25.0	45	60	10	0	23.3	30
w/o Keyframe-forecasting		90	45	25	53.3	80	70	50	30	50.0	75
w/o Textual-planning		75	40	15	43.3	70	65	25	10	33.3	50
BagelVLA		95	65	60	73.3	95	80	65	45	63.3	85

Long-Horizon Task

Real-world

Dual Flow-Matching Schemes	Latency↓	ABC-D↑
Complete Denoise	6.04s	2.480
Joint Denoise	2.90s	2.038
Single-step Denoise (Eq. 2)	1.23s	3.345
RFG (Eq. 3)	1.23s	3.600 ?



RFG for Action Generation + Physical Constraints

Attention-Guided Sampling in Gaussian Latent Manifold



西北工业大学
NORTHWESTERN POLYTECHNICAL UNIVERSITY

Thanks for listening!

Zhengwei Yang

2026.05.31